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Chapter 1

Boats

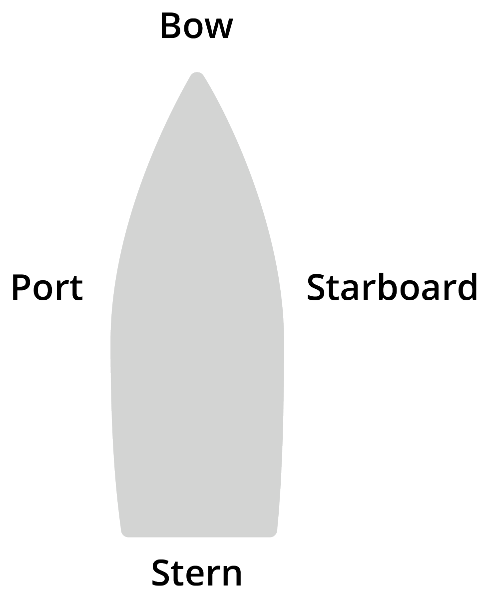
Engineers have been building boats for centuries. Through boat design, humanity learned the lessons that made airplanes and rockets possible. Learning about how boats work will give you a foundation for understanding more advanced concepts we will be covering later.

1.1 Basic Terminology

The front of a boat is called *the bow* (pronounced exactly the same as ”bough”). The back of the boat is called *the stern*.

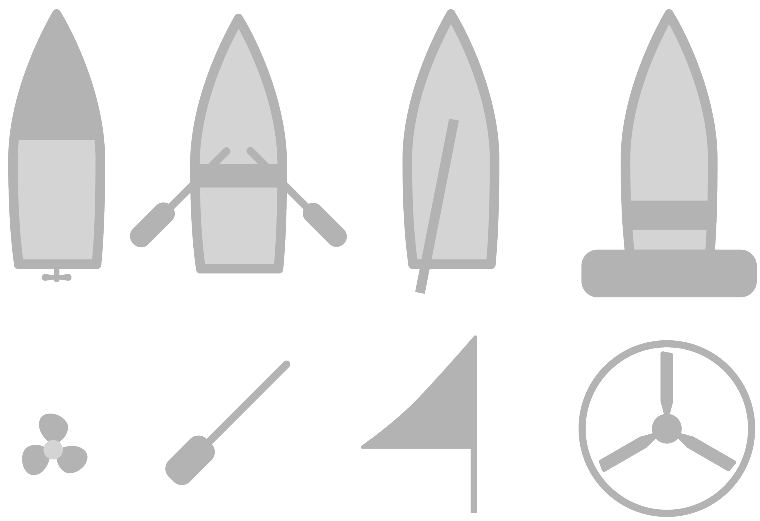
The underside of the boat is called *the hull*. The top of the boat is called *the deck*.

If you are standing at the stern and looking toward the bow, everything on your left is the *port* side. Everything on your right is the *starboard* side.



There are several different ways that boats are propelled:

- A motor turns a screw in the water, as in a motorboat. The screw is known as a *propeller*.
- A human pushes the water with a big spatula. If the spatula is attached to the boat with a pivot (as in a rowboat) it is an *oar*. If the spatula is not attached to the boat (as in a canoe), it is a *paddle*.
- A human pushes the ground beneath the water with a long pole. These boats are called *punts*, and are mainly used in more shallow waters where an oar or a paddle would be less convenient.
- The wind pushes the boat, as in a sailboat. The sails are held up by a *mast*.
- Some boats have a big fan that propel the boat. These are called *airboats*. Airboats are not the most efficient boats, but they can travel on waterways with water just a couple of inches deep.



In the terms of physics, each of these method provide a *thrust vector* which is applied to the boat at a particular place and in a particular direction.

1.1.1 Nautical Miles and Knots

A *nautical mile* is use to measure distances on a boat. The speed of a boat is usually measured in *knots*, which is just a quick way of saying ”nautical miles per hour”.

What is a nautical mile?

Latitude is the angle that a boat is above or below the equator. It is measured in degrees, minutes, and seconds. 60 minutes make one degree. 60 seconds make one minute. So, for example, if you are 12.45164° north of the equator, a sailor might say you were at Latitude 12°27′5"N. She might also say that you are a little more than 747 nautical miles from the equator. $(12 \times 60 + 27 = 747)$

A nautical mile is the distance of one minute of latitude. It is slightly longer than the normal (or *statute*) that you use on land: one nautical mile equals 1.151 statute miles or 1.852 km per hour.

1.2 Why Boats Float Upright

Early in this sequence, we discussed buoyancy as a quantity. The magnitude of the buoyant force is equivalent to the weight of the liquid displaced.

We can also talk about the direction of the buoyant force: buoyancy pushes in the opposite direction as gravity.

How do we design boats so that they don't flip over?

1.2.1 Center of Buoyancy

Let's say you have a rowboat. If you push down on point on the floor near the front, the front of the boat will go down and the back of the boat will rise. In other words, the boat will rotate in the direction you are pushing. Likewise, if you push on the floor near the back of the boat, the back will sink a little lower and the front will rise. However, there is a place, near the center of the boat, where if you push down, the boat will not rotate at all; it will simply sink a little lower in the water. That point is known as the *center of buoyancy*.

How can we calculate the center of buoyancy? Imagine the shape of the water that was displaced by the boat. Now imagine that shape filled with water. The center of mass of that water is the center of buoyancy of the boat.

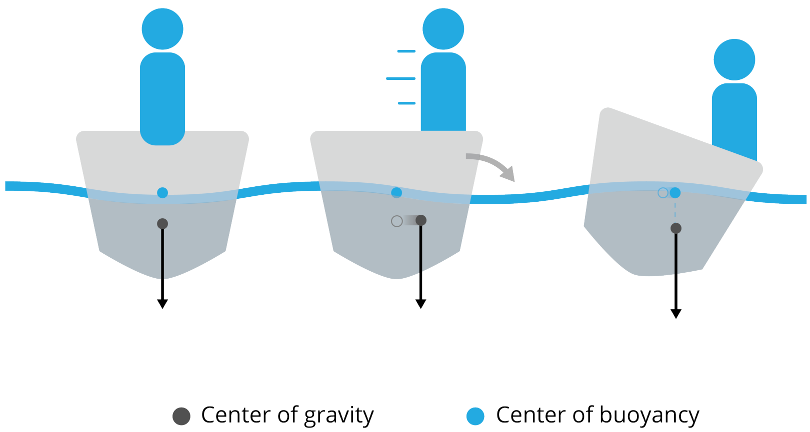
1.2.2 Center of Mass

Your boat and everything in it can be thought of as one object. That object has a center of mass. If you found the center of mass, you could balance the whole boat on it.

In a boat, if you move your body from the center of the boat to once side, you will have moved the center of mass. The boat will lean in that direction, which will change the center of buoyancy.

If you imagine a line is parallel to the force of gravity that passes through the center of mass of your boat, the boat will continue to increase its lean until the center of buoyancy is on that line.

If water comes over the sides of the boat before the center of gravity and center of buoyancy align, your boat will sink.



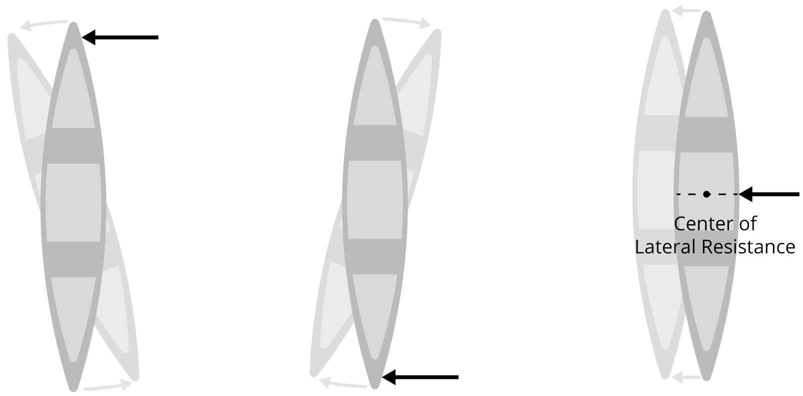
1.3 Center of Lateral Resistance

It isn’t enough for a boat to float — for a boat to be useful, it must also be able to travel in a straight line.

Imagine that you are standing knee-deep in a lake next to a canoe. If you push the front of the canoe away from you, it will rotate — the back end will actually swing toward you. There is a point near the middle of the canoe where if you push it will not rotate in either direction — the boat will just slide sideways. This point is known as the *center of lateral resistance*.

The trick to making a boat travel in a straight line is to make sure that the line that contains the thrust vector passes through the center of lateral resistance.

An outboard motor allows you to direct the thrust vector. When the line of thrust passes the center of lateral resistance on the starboard side of the center of lateral resistance, the boat turns toward the port side.

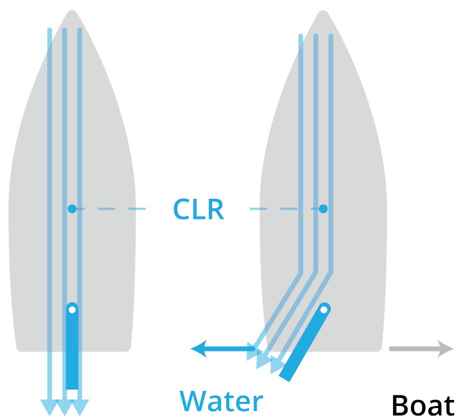


1.4 Steering with a Rudder

While outboard motors and airboats let you direct the thrust vector, most boats have a *rudder*. The rudder is a blade on a pivot near the back of the boat. The angle of the rudder can be adjusted so that water rushing past it gets pushed to one side or the other.

According to Newton's third law, when the water gets pushed to the left, the back of the boat gets pushed (with the same force) to the right. This causes the boat to rotate around its center of lateral resistance.

Note that a rudder only works when the boat is passing through the water.



1.5 Boat Length and Resistance

FIXME: Write about wave length, boat length, and Froude number.

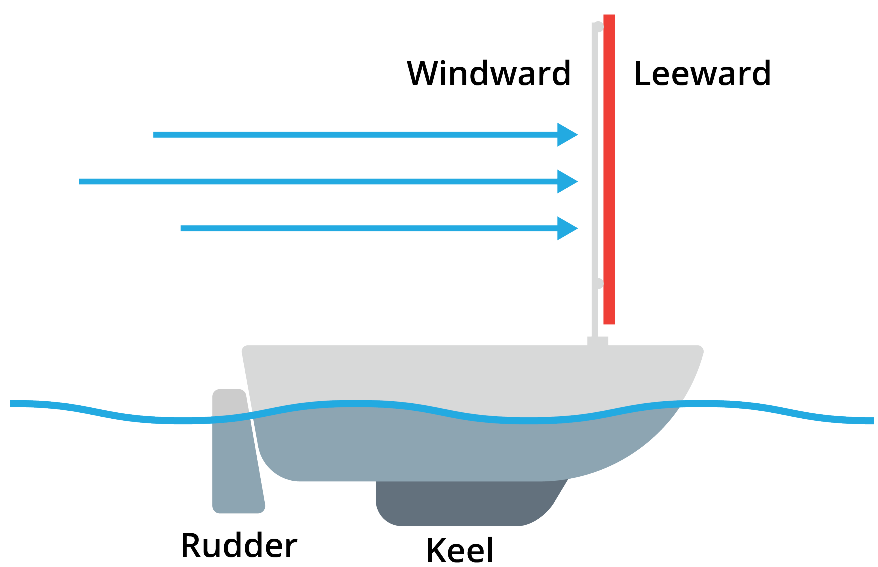
Chapter 2

Sailboats

Imagine that you have a canoe, and you are about to paddle from one island to another that is directly east of where you are standing. There is also a steady wind coming from the west, and you have a big piece of plywood. You might be inspired to use it as a sail.

This situation is the most simple form of sailing: Wind comes from behind the boat and hits the sail which generates a force that pushes the boat in the direction of the wind.

The sail has two sides: The *windward* side is the one that is getting hit with the wind. The *leeward* side is the side away from the wind.



2.1 Magnitude of the Wind Force

The first natural question is: How much force will I have pushing my canoe through the water?

Wind Force

When the sail is perpendicular to the wind, the force of the wind on the sail in newtons (F_w) will be given by:

$$F_w = A\frac{dv^2}{2}$$

where A is the area of the sail in square meters d is the density of the gas in kg per cubic meter, and v is the wind speed in meters per second.

For air at STP, d is about 1.225 kg per cubic meter.

We call $\frac{dv^2}{2}$ the *wind pressure*. This is the amount of pressure that the windward side of plywood is experiencing that is above the pressure that the leeward side of the plywood is experiencing. (The leeward side might experience some turbulence, but the pressure it is experiencing is approximately 1 atmosphere.)

Let's say your canoe is standing still and the wind is 0.5 m/s. This means the wind pressure is

$$P = \frac{1.225(0.5^2)}{2} = 0.153125 \text{ newtons per square meter}$$

Let's say your plywood sail is 2 meters tall and 1.5 meters wide. What will be the force of the wind?

$$F_w = AP = (3)(0.153125) \approx 0.46 \text{ newtons}$$

This is a very intuitive idea: There is a difference between the pressure on the windward side and the pressure on the leeward side, and the plywood experiences a force that pushes the boat through the water.

2.2 Direction and Location of the Wind Force

If there is low pressure on one side of the sail and high pressure on the other, the force vector will be perpendicular to the sail.

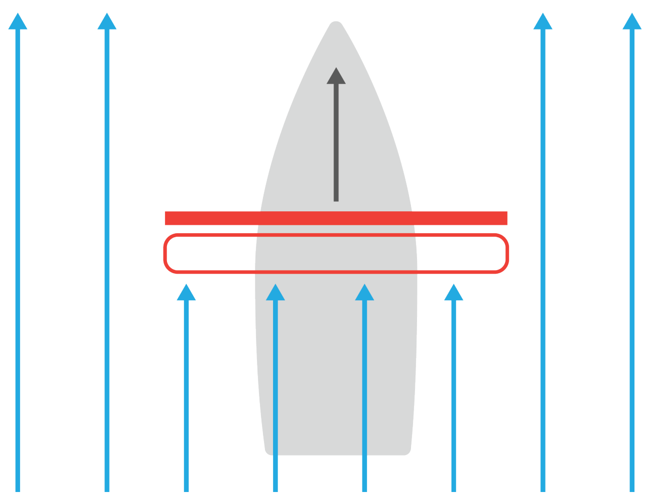
Where is this force vector applied? We can think of the force as being applied at the geometric center of the sail. This is called *the center of effort*. In this case, the center of effort is the exact center of the rectangular plywood.

The mast on a windsurfing board can be tilted from side to side. When the center of effort is over the center of lateral resistance, the board goes straight. To steer, the sailor moves the mast to one side of the center of lateral resistance, which rotates the board.

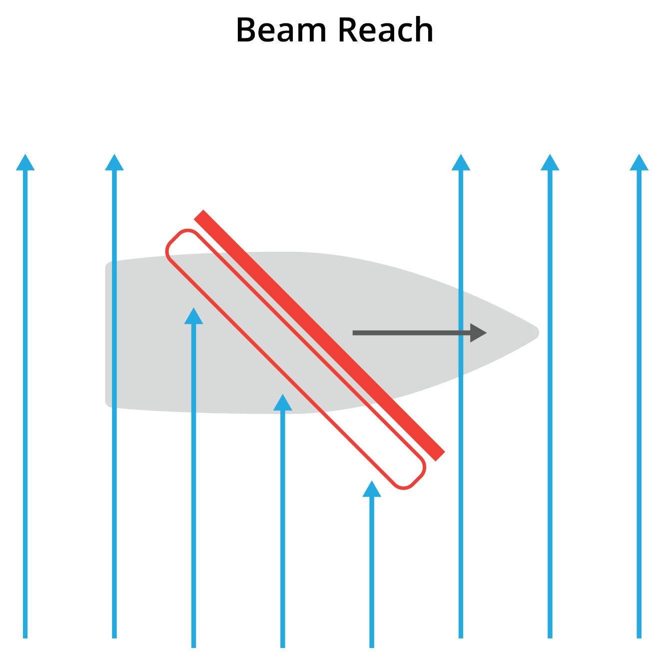
2.3 Beam Reach

When you are sailing in the same direction as the wind, sailors say you are *running*.

Running



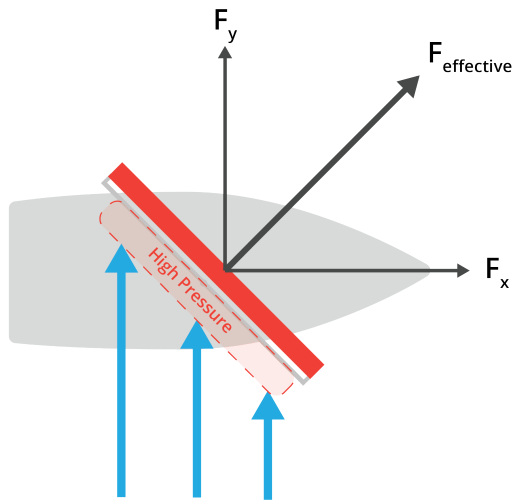
What if you want to go east and the wind (still 0.5 m per second) is now coming from the south? Sailing perpendicular to the wind is known as a *beam reach*.



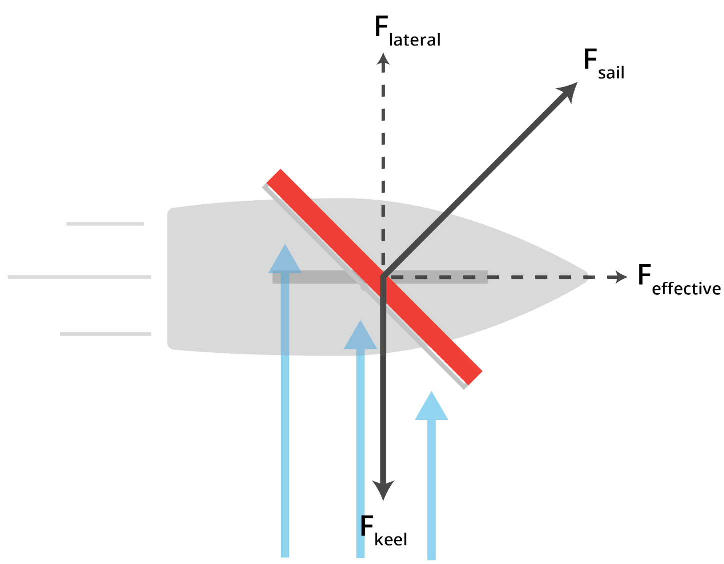
To do a beam reach, instead of mounting the plywood perpendicular to the boats direction of travel, you would mount it at a 45 degree angle. The wind pressure will build on the windward side of the plywood, and the plywood will experience a force pushing it at a 45° angle to the boat.

We can think of this force as having two components:

- One component pushes the boat forward (Yay!)
- One component pushes the boat sideways (Ugh!)



To minimize the effect of the sideways force, sailboats typically have a keel — a long fin on its underside that slows its sideways sliding.



Notice also that the "wind shadow" of the plywood is smaller when it is at a 45° angle to the wind. How much smaller? The effective area of your plywood has gone from 3 square meters to $\frac{3}{\sqrt{2}} \approx 2.12$ square meters. This means the force generated will be smaller, and some of it will be wasted pushing the boat sideways.

If we assume that the wind pressure is still 0.153125 newtons per square meter. The force on the plywood will be about

$$F_w = AP = \frac{3}{\sqrt{2}}(0.154125) \approx 0.325 \text{ newtons}$$

However, the direction of that force is not all in the direction you want to go, so the effective force is

$$F = F_w \frac{1}{\sqrt{2}} = \frac{3}{2}(0.154125) = 0.2311875 \text{ newtons}$$

Notice that we got twice as much effective force when we were running with the wind as when we are on a beam reach. However, any sailor will tell you that you can go much faster on a beam reach than you can running. Why?

2.4 Apparent Wind

When you are running, you can never go faster than the wind. As you go faster and faster, the wind that the boat experience decreases. For example, if you are going 0.2 m/s in a wind of 0.5 m/s, you (and your sail) will only experience wind at 0.3 m/s. We call the wind as experienced by the boat the *apparent wind*. The wind as observed by a stationary observer is called the *actual wind*.

If you are running with the wind, as you approach the speed of the wind, the force of that wind will decrease towards zero.

On a beam reach, as you go faster, the direction of the wind seems to change. If you are going 0.2 m/s east and the actual wind is 0.5 m/s from the south, the direction of the apparent wind will seem to come from about 22 degrees east of true south. The speed of the apparent wind will be about 0.54 m/s.

2.5 Close Reach

What if you want to go east, and the wind is coming from 40 degrees east of south? This would mean that you were sailing just 50 degrees away from straight into the wind. Is this possible?

If you put your sail at a 25 degree angle, you will still catch some wind and create some pressure on one side of the sail. Most of the resulting force would be trying to push your boat sideways, but some of it would be in the direction you were trying to travel.

Picking an angle for your sail that creates high pressure that makes a desirable force is known as the ”angle of attack”.

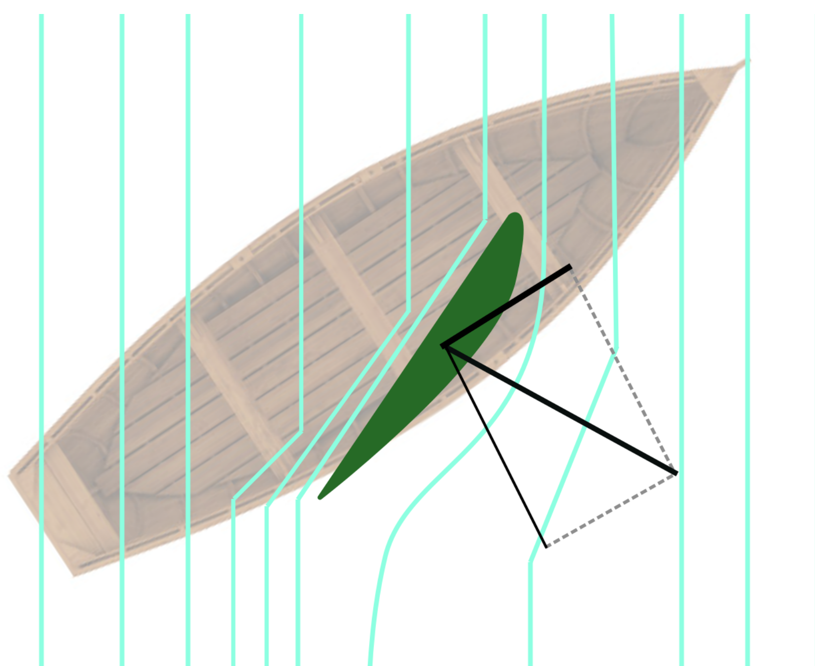
This is a result does not feel intuitive. A boat can sail into the wind!? The boat can’t sail directly into the wind – with each degree that the boat gets closer to straight into the wind, the force pushing it forward decreases and the force pushing it back increases. However, most boats can get within 45% if they have a well-shaped sail.

2.6 Shaping the Sail

Most of the power of the wind can be captured with a piece of flat plywood. The wind hits it and creates a high pressure on the windward side. What about the other side of the plywood?

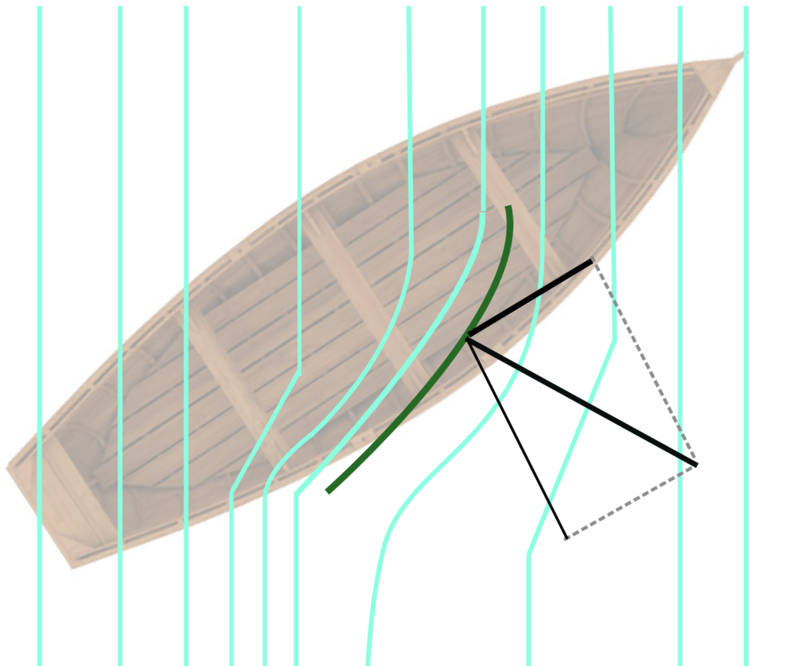
It turns out that if we can get the wind to travel smoothly over the back side of the plywood, the pressure on that side will be a little lower than if we had turbulence there. (We are not going to go too deeply into why. If you want to learn more, look up the Coanda effect.)

For example, if we were on a close reach, the very best sail we could have would gently pull the wind along its backside. It would look like this:



Of course, for the sail to work on either side of the boat, this asymmetrical design would not work. (Although, we should note that this design works great for airplane wings.)

When we make a sail out of cloth, we give it some curve known as *camber*. Slow winds require just a little camber; fast winds require more.



Some newer sailboats have wing sails that have two pieces that can be arranged to redirect the most air possible.

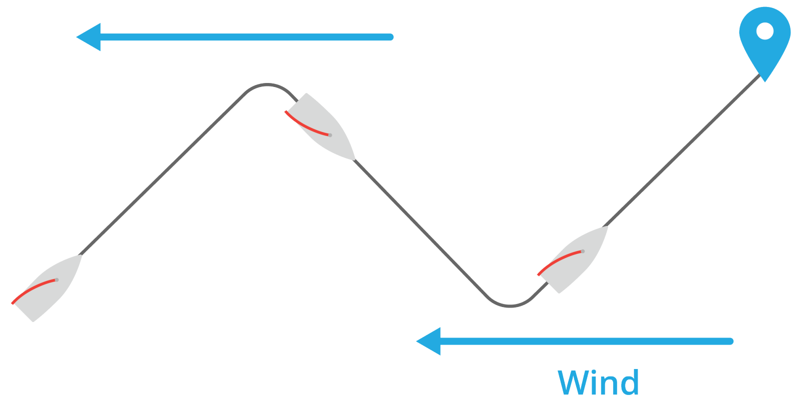
Note: when running with the wind, the turbulence on the leeward side of the sail is unavoidable. But when traveling perpendicular to the wind or on a close reach, the air should move smoothly over the leeward side of the sail. Many sailors have a piece of yarn taped on each side of the sail so they can see if the air is moving smoothly.

Many sailboats also have multiple sails. Besides the increase in the sail area, each sail also redirects the wind to pass smoothly over the leeward surface of the sail behind it.

2.7 Tacking into the Wind

What if the wind is coming from the east and you really need to go directly into the wind?

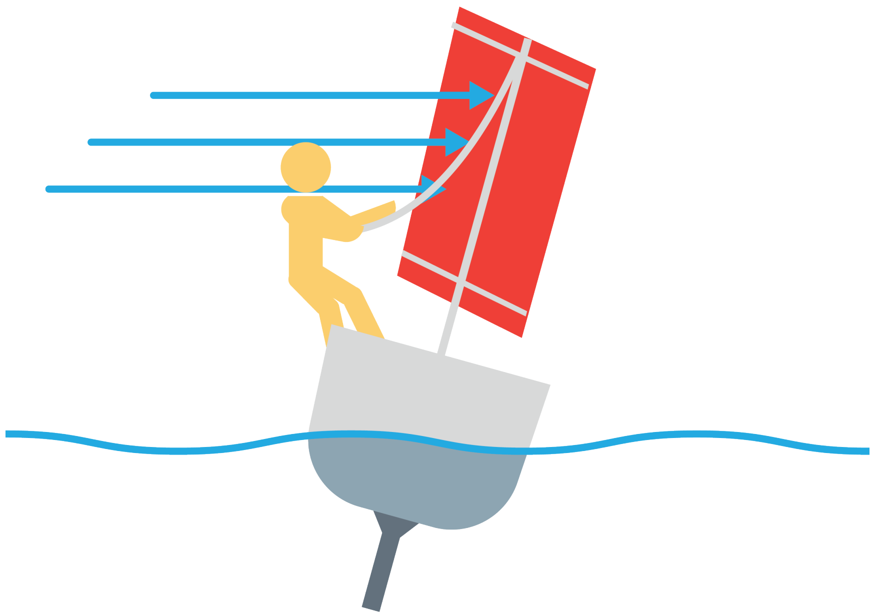
The boat will not travel straight into the wind, instead you will travel on a close reach with the wind coming from one side of the boat. You will then turn into the wind and continue turning until you are on a close reach with the wind coming from the other side. This is known as *tacking*.



2.8 Heeling

The center of effort is near the center of your sail, which is usually pretty high in the air. Yes, the force generated will push your boat, but it will also rotate your boat. Sailors call the resulting lean *heeling*. Heeling too far is problematic — the rudder gets pulled out of the water, which makes it hard to steer the boat.

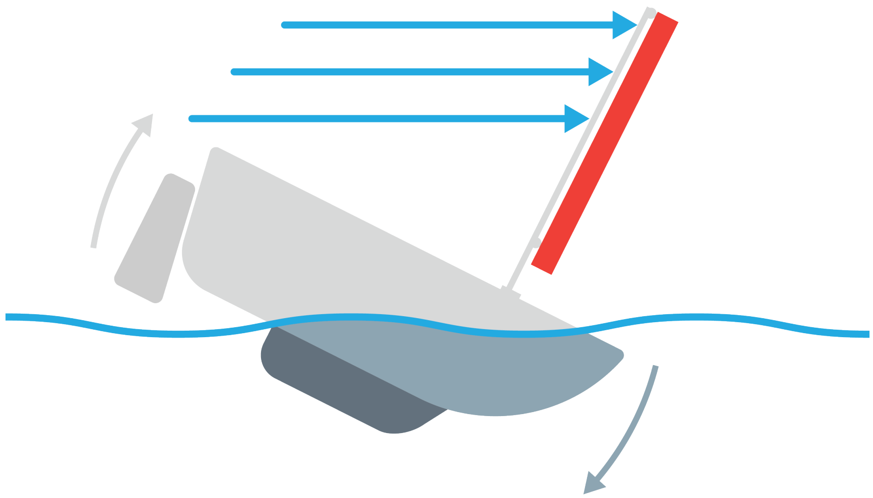
If a boat is heeling too far, sailors will move their weight to the windward side of the boat — some even wear harnesses that push their weight out beyond the edge of the hull. If that doesn't work, they will reduce the sideways force on the sail.



Can the boat flip over? (The word sailors use is *capsize*.) Most larger boats should not be able to capsize; as the sail gets pushed down toward the water, it loses power. So, putting some weight in the keel will ensure that the boat doesn't turn upside-down. Small boats (think 3 or 4 meters long) can capsize, but they are small enough that the sailor can usually get them upright again without assistance.

There are boats with multiple hulls. A catamaran has two identical hulls that are side-by-side. As a result, the catamaran will not heel much at all. However, if a big gust of wind comes up, one hull can be pulled completely out of the water. Catamarans can be capsized.

If a boat is running when a big gust comes up, that rotating force will try to push the bow underwater. If the boat is going very fast, this can result in a somersault as the front of the boat slows and dips suddenly and the back of the boat is pitched up over it.



Chapter 3

Airplanes

Now that you understand how a sail works, you are ready to understand how an airplane works.

3.1 Drag and Thrust

As an airplane flies, the air around it creates a large force that is trying to slow the plane down. Pilots call this *drag*. To overcome drag, an airplane must have a source of *thrust*. This is usually done by pushing air toward the back of plane. You can think of drag as air resistance or air friction, and thrust as the applied force overcoming that friction.

There are three basic types of propulsion on airplanes:

- Propeller planes have fan blades that are spun by an internal combustion engine and push the air toward the back of the plane.
- Jet planes suck air into a tube, mix it with fuel, and ignite the mixture. There are fan blades inside the tube that ensure the power from the burn is efficiently converted into thrust.
- Turboprop planes use both propeller and jet technology — their propellers are turned by jet engines.

Drag is proportional to the square of the velocity. For example, if you double how fast you are flying, the drag force will increase by a factor of 4.

Passenger airplanes fly fast — they have no trouble reaching 1000 km per hour. A Boeing 747 burns about 4 liters of fuel per second; the flight from London to New York requires about 70,000 kg of fuel. When the plane takes off, the fuel for the journey often weighs twice as much as the passengers.

3.2 Lift

Unlike a hot air balloon, an airplane is heavier than the air it displaces, so the force of gravity will pull it from the sky unless there is a counteracting force. We call the counteracting force *lift*. Lift works just like sailing into the wind. By picking an angle of attack, we create high pressure under the wing. By creating a nice smooth path for the air to travel over the top of the wing, we create low pressure over the wing. This difference creates the lift on the wings. Wings are shaped in this way to make it easier for the air to flow over the top of the wing, a shape called an *airfoil*. An airfoil is any shape that produces more lift than drag, such as a kite, a blade, an airplane wing, or sail.

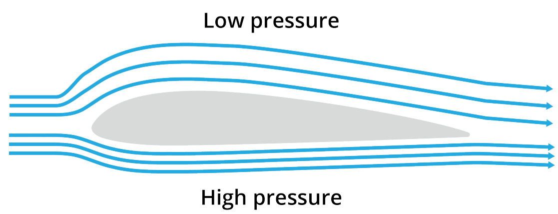


Figure 3.1: A diagram of the flow of air on a wing cross section.

(There are some very bad explanations of this idea that overstate the importance of the rounded top of the wing. Most airplanes can fly upside down — if the most important part were the shape of the top, this would be impossible. Instead, the most important part is the angle of attack.)

At high altitudes, the air is less dense, until space is reached where there is no air and it is an entire vacuum. If two identical planes are flying at the same speed, but at different altitudes:

- There is less drag on the higher airplane.
- The wings provide less lift on the higher airplane.
- The air around the higher plane has less oxygen per liter, which can affect how the fuel burns.

3.3 Control: Roll, Pitch, Yaw

The first control that a pilot has is the throttle. By increasing the throttle, the pilot increases the plane's thrust.

The pilot has a stick. Pulling back on the stick causes the *elevator* on the tail of the plane to go up. Air hitting the elevator pushes the tail down and the nose up.

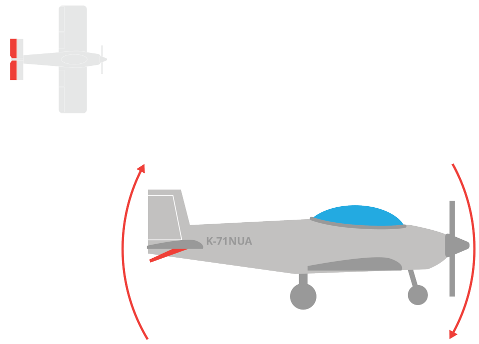


Figure 3.2: A diagram of a top-down and side view of an elevator and its pitch effect

Pushing the stick forward will push the tail up and the nose down. We say ”The elevator controls the *pitch* of the airplane.”

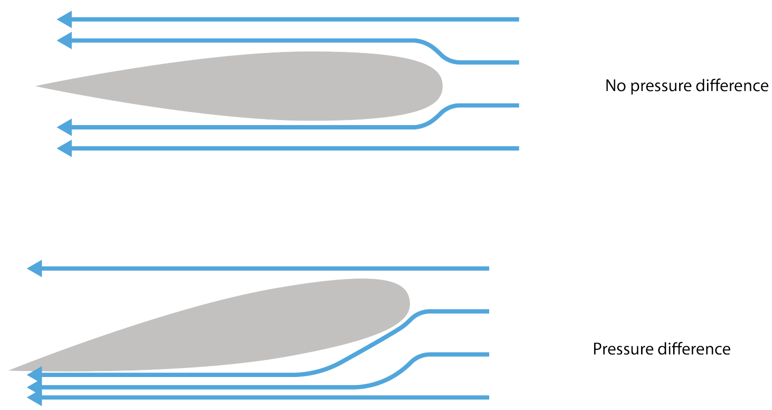


Figure 3.3: A view of how the wing of the airplane is effected by pressure.

The stick also goes left and right. Pushing the stick to the left, lifts the *aileron* on the left wing and lowers the aileron on the right wing. This pushes the left wing down and the right wing up. We say “The ailerons control the *roll* of the airplane”.

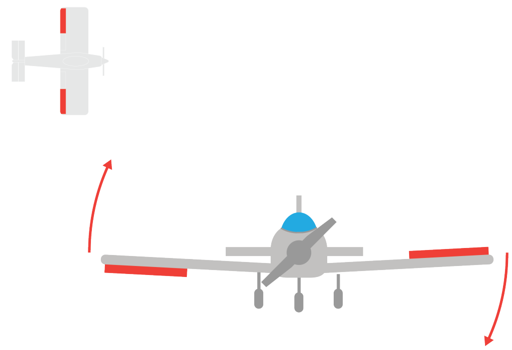


Figure 3.4: Aileron locations on the plane and their effect on roll.

The pilot controls the *rudder* on the tail with his feet. Pushing the right side down will push the rudder to the right. This will push the tail of the plane to the left and the nose to the right. We say "The rudder controls the *yaw* of the airplane."

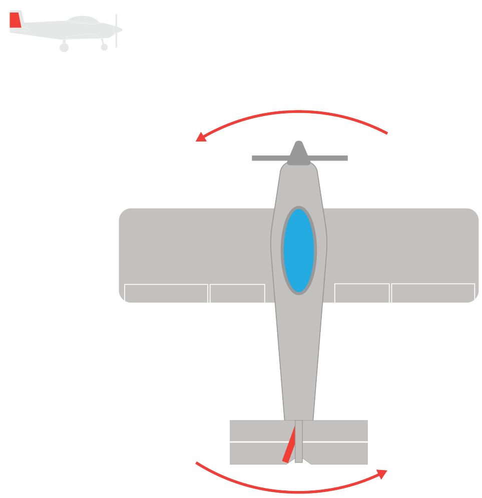


Figure 3.5: The rudder controls the yaw of the airplane, or top-down rotation

The flaps of a plane are surfaces on edges of the underside of the wings that are used to increase lift and drag during takeoff or landing.

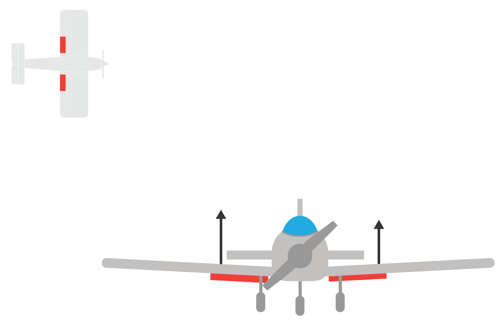


Figure 3.6: A diagram showing the location and effects of the flaps.

3.4 Thrust

There are several common ways that airplanes produce thrust.

The first is a propeller powered by a standard piston engine.

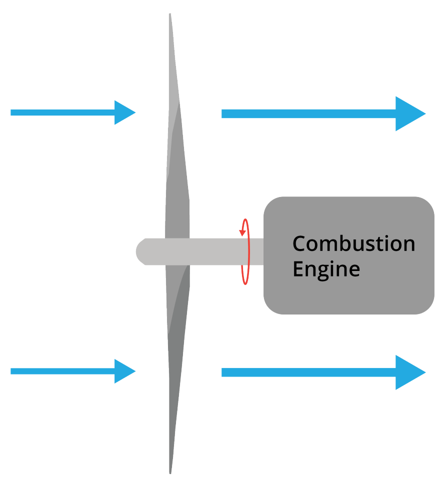


Figure 3.7: The propeller uses the air to push it forward (thrust). The wings of a propeller act similar to sails and wings of an airplane, separating airflow.

Another way to produce thrust is by using a *jet engine*. Jet engines take in air, increase its pressure and temperature, and shoots it out of the back of the engine.

Jet engines suck in air, compress it, add a fuel mixture, ignite the fuel, and shoot it out the back. Once the air exits the engine, it expands at a high speed, which further helps create thrust.

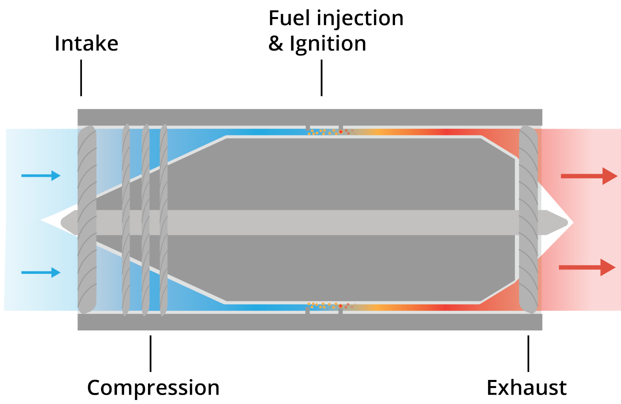


Figure 3.8: Cross section of a jet engine

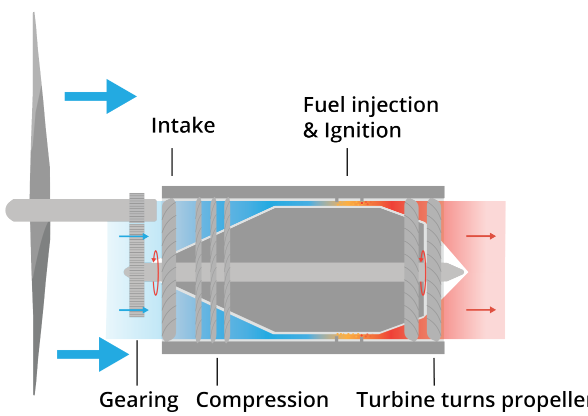


Figure 3.9: Cross section of a turboprop

3.5 Gliders

It is interesting to note that gliders fly without an engine. Gliding usually starts high in the air, where the glider has lots of potential energy. To stay aloft for a long time, glider pilots will look for places where air is rising, so they can ride those updrafts and regain that potential energy.

Chapter 4

Data Tables and pandas

Much of the data that you will encounter in your career will come to you as a table. Some of these tables are spreadsheets, some are in relational databases, and some will come to you as CSV files.

Typically, each column will represent an attribute (like height or acreage) and each row will represent an entity (like a person or a farm). You might get a table like this:

`property_id`	`bedrooms`	`square_meters`	`estimated_value`
7927	3	921.4	$ 294,393
9329	2	829.1	$ 207,420

In most cases, one of the columns is guaranteed to be unique. We call this the *primary key*. In this table, `property_id` is the primary key; every property has one, and no two properties have the same `property_id`.

4.1 Data types

Each column in a table has a type, and these usually correspond pretty nicely with types in Python.

Here are some common datatypes:

Type	Python type	Example
Integer	int	`910393`
Float	float	`-23.19`
String	string	`'Fred'`
Boolean	bool	`False`
Date	datetime.date	`2019-12-04`
Timestamps	datetime.datetime	`2022-06-10T14:05:22Z`

Sometimes it is OK to have values missing. For example, if you had a table of data about employees, maybe one of the columns would be `retirement` — a date that tells you when the person retired. People who had not yet retired would have no value in this column. We would say that they have *null* for `retirement`.

Sometimes there are constraints on what values can appear in the column. For example, if the column were `height`, it would make no sense to have a negative value.

Sometimes a column can only be one of a few values. For example, if you ran a bike rental shop, each bicycle's status would be "available", "rented", or "broken". Any other values in that column would not be allowed. We often call these columns *categorical*.

4.2 pandas

The Python community works with tables of data *very* often, so it created the pandas library for reading, writing, and manipulating tables of data.

When working with tables, you sometimes need to go through them row-by-row. However, for large datasets, this is very slow. pandas makes it easy (and very fast) to say things like "Delete every row that doesn't have a value for height" instead of requiring you to step through the whole table.

In pandas, there are two datatypes that you use a lot:

- a `Series` is a single column of data.
- a `DataFrame` is a table of data: it has a `Series` for each column.

In the digital resources, you will fined `bikes.csv`. If you look at it in a text editor, it will start like this:

```
bike_id,brand,size,purchase_price,purchase_date,status
5636248,GT,57,277.99,1986-09-07,available
4156134,Giant,56,201.52,2005-01-09,rented
7971254,Cannondale,54,292.25,1978-02-28,available
3600023,Canyon,57,197.62,2007-02-15,broken
```

The first line is a header; it tells you the name of each column. Next, the each bike is represented by values which are separated by commas. (Thus the name: CSV stands for "Comma-Separated Values".)

4.3 Reading a CSV with pandas

Let's make a program that reads `bikes.csv` into a pandas dataframe. Create a file called `report.py` in the same folder as `bikes.csv`.

First, we will read in the csv file. pandas has one series that acts as the primary key; it calls this one the index. When reading in the file, we will tell it to use the `bike_id` as the index series.

If you ask a dataframe for its shape, it returns a tuple containing the number of rows and the number of columns. To confirm that we have actually read the data in, let's print those numbers. Add these lines to `report.py`:

```
import pandas as pd

# Read the CSV and create a dataframe
df = pd.read_csv('bikes.csv', index_col="bike_id")

# Show the shape of the dataframe
(row_count, col_count) = df.shape
print(f"*** Basics ***")
print(f"Bikes: {row_count:,}")
print(f"Columns: {col_count}")
```

Build it and run it. You should see something like this:

```
*** Basics ***
Bikes: 998
Columns: 5
```

Note that your table actually has six columns. The index series is not included in the shape.

4.4 Looking at a Series

Let's get the lowest, the highest, and the mean purchase price of the bikes. The purchase price is a series, and you can ask the dataframe for it. Add these lines to the end of your program:

```
# Purchase price stats
print("\n*** Purchase Price ***")
series = df["purchase_price"]
print(f"Lowest:{series.min()}")
print(f"Highest:{series.max()}")
print(f"Mean:{series.mean():.2f}")
```

Now when you run it, you will see a few additional lines:

```
*** Purchase Price ***
Lowest:107.37
Highest:377.7
Mean:249.01
```

What are all the brands of the bikes? Add a few more lines to your program that shows how many of each brand:

```
# Brand stats
print("\n*** Brands ***")
series = df["brand"]
series_counts = series.value_counts()
print(f"{series_counts}")
```

Now when you run it, your report will include the number of bikes for each brand from most to least common:

```
*** Brands ***
Canyon        192
BMC           173
Cannondale    170
Trek          166
GT            150
Giant         147
Name: brand, dtype: int64
```

`value_counts` returns a Series. To format this better, we need to learn about accessing individual rows in a series.

4.5 Rows and the index

In an array, you ask for data using the location (as an int) of the item you want. You can do this in pandas using `iloc`. Add this to the end of your program:

```
# First bike
print("\n*** First Bike ***")
row = df.iloc[0]
print(f"{row}")
```

When you run it, you will see the attributes of the first row of data:

```
*** First Bike ***
brand                    GT
size                     57
purchase_price       277.99
purchase_date    1986-09-07
```

```
status                available
Name: 5636248, dtype: object
```

Notice that the data coming back is actually another series.

The last line says that the name (the value for the index column) for this row is 5636248. In pandas, we usually use this to locate particular rows. For example, there is a row with `bike_id` equal to 2969341. Let's ask for one entry from the

```
print("\n*** Some Bike ***")
brand = df.loc[2969341]['brand']
print(f"brand = {brand}")
```

Now, you will see the information about that bike:

```
*** Some Bike ***
brand = Cannondale
```

pandas has a few different ways of getting to that value. All of these get you the same thing:

```
brand = df.loc[2969341]['brand'] # Get row, then get value
brand = df['brand'][2969341]      # Get column, then get value
brand = df.loc[2969341, 'brand'] # One call with both row and value
```

4.6 Changing data

One of your attributes needs cleaning up. Every bike should have a status, and it should be one of the following strings:"available", "rented", or "broken". Get counts for each unique value in status:

```
print("\n*** Status ***")
series = df["status"]
missing = series.isnull()
print(f"{missing.sum()} bikes have no status.")
series_counts = series.value_counts()
for value in series_counts.index:
    print(f"{series_counts.loc[value]} bikes are \"{value}\"")
```

This will show you:

```
*** Status ***
7 bikes have no status.
389 bikes are "rented"
304 bikes are "broken"
296 bikes are "available"
1 bikes are "Flat tire"
1 bikes are "Available"
```

Right away, we can see two easily fixable problems: Someone typed “Available” instead of “available”. Right after you read the CSV in, fix this in the data frame:

```
mask = df['status'] == 'Available'
print(f"{mask}")
df.loc[mask, 'status'] = 'available'
```

When you run this, you will see that the mask is a series with `bike_id` as the index and `False` or `True` as the value, depending on whether the row’s status was equal to “Available”.

When you use `loc` with this sort of mask, you are saying “Give me all the rows for which the mask is True.” So, the assignment only happens in the one problematic row.

Let’s get rid of the mask variable and do the same for turning `Flat tire` into `Broken`:

```
df.loc[df['status'] == 'Available', 'status'] = 'available'
df.loc[df['status'] == 'Flat tire', 'status'] = 'broken'
```

Now those problems are gone:

```
7 bikes have no status.
389 bikes are "rented"
305 bikes are "broken"
297 bikes are "available"
```

What about the rows with no values for status? If we were pretty certain that the bikes were available, we could just set them to ’available’:

```
missing_mask = df['status'].isnull()
df.loc[missing_mask, 'status'] = 'available'
```

Or maybe we would print out the IDs of the bikes so that we could go look for them:

```
missing_mask = df['status'].isnull()
missing_ids = list(df[missing_mask].index)
print(f"These bikes have no status:{missing_ids}")
```

However, let's just keep the rows where the status is not null:

```
missing_mask = df['status'].isnull()
df = df[~missing_mask]
```

At the end of your program, write out the improved CSV:

```
df.to_csv('bikes2.csv')
```

Run the program and open `bikes2.csv` in a text editor.

4.7 Derived columns

Let's say that you want to add a column with age of the bicycle in days:

```
bike_id,brand,size,purchase_price,purchase_date,status,age_in_days
5636248,GT,57,277.99,1986-09-07,available,13061
4156134,Giant,56,201.52,2005-01-09,rented,6362
7971254,Cannondale,54,292.25,1978-02-28,available,16174
```

Your first problem is that the `purchase_date` column looks like a date, but really it is a string. So, you need to convert it to a date. You can do this by applying a function to every item in the series:

```
df['purchase_date'] = df['purchase_date'].apply(lambda s:
    datetime.date.fromisoformat(s))
```

(With pandas, there is often more than one way to do things. pandas has a `to_datetime` function that converts every entry in a sequence to a `datetime` object. Here is another way to convert the string column in to a date column:

```
df['purchase_date'] = pd.to_datetime(df['purchase_date']).dt.date
```

You can look up `dt` and `date` if you are curious.)

Now, we can use the same trick to create a new column with the age in days:

```
today = datetime.date.today()
df['age_in_days'] = df['purchase_date'].apply(lambda d: (today - d).days)
```

When you run this, the new `bikes.csv` will have an `age_by_date` column.

Chapter 5

Rotorcraft

5.1 Introduction

This chapter moves on from airplane movement to quadcopters! You may here rotor-winged aircraft, drones, or unmanned aerial vehicle, which all refer to similar types of the same structure.

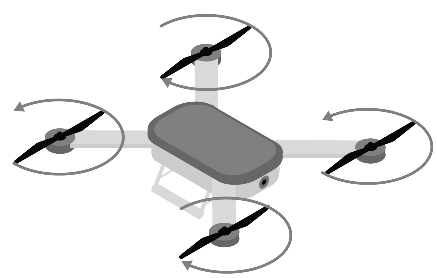


Figure 5.1: The general look of a quadcopter. Includes four rotors, a frame called a fuselage, and generally a head or designated front.

Before we dive in depth, let's introduce a few terms that will be repeated within this chapter.

Quadcopter Often defined as an aircraft with 4 rotors, synonymous with drone.

Rotor The rotor is the part of the aircraft which causes lift by spinning at a high velocity. It consists of a central *hub* with two or more *blades* attached. You may hear the term propeller like from our airplanes chapter, but the more correct term for quadcopters is *rotor*.

UAS / UAV Unmanned Aircraft System or Unmanned Aircraft Vehicle. Often used to describe drones or semi-autonomous quadcopters. This includes the drones themselves, the software needed to avoid obstacles and track positioning, and equipment used by people to communicate with them.

Fuselage The main body of a quadcopter or aircraft; synonymous to frame.

Torque We have not talked about torque yet, but it is an important circular motion concept. Torque is the rotational equivalent of force, and is a measure of how much a force acting on an object causes that object to rotate. For now, understand that it is affects quadcopter movement and is a force that needs to be counteracted in order to maintain stability.

Isometric We also want to be clear about our use of views in our diagrams. An *isometric* perspective is one that visually represents all 3 views of a 3D object in a 2D space (a printed diagram). When we get to calculus, we will talk about *isometry* as transformation or mapping between two spaces that perfectly preserves distances.

5.2 VTOL

The strength of quadcopters lies in their propulsion system. As compared to airplanes, which take off by accelerating down *runways* and gaining enough lift to overcome its weight, quadcopters use spinning rotors to generate this lift. This applies Newton's Third Law — for every action (the downwards lift of the rotors), there is an equal and opposite reaction (the upwards lift of the quadcopter). This allows quadcopters to take off vertically, eliminating the need for a runway. This is a very special type of propulsion known as Vertical Take-Off and Landing (VTOL).

Specifically, the drone's rotors needs to generate enough lift to counteract its own weight and put it into motion in the air. Recall that there are *equal but opposite* forces in play here, so the lift created by the rotors can be found by

$$F_{\text{rotors}} = m_{\text{quadcopter}}\, g + F_{\text{vertical lift}}$$

This is a rudimentary equation that demonstrates the following: the amount of lift created by rotors is the excess force after the rotors are able to overcome the weight of the quadcopter.

Here is an isometric view of vertical flight of a quadcopter

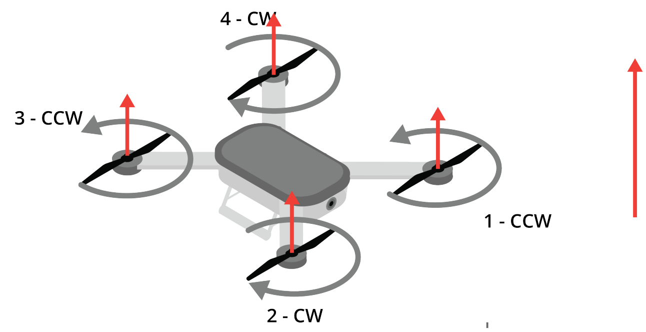


Figure 5.2: A quadcopter taking vertical flight. Notice how the red arrows, representing the strength of the upwards force, are all equivalent.

If we just want our quadcopter to *hover in place*, the forces vertically need to be equal. This means that the weight of the quadcopter will equal the lift generated by the rotors

$$F_{\text{rotors}} = m_{\text{quadcopter}}\, g \tag{5.1}$$

5.3 Rotor Structure

Now, the remaining question is how do rotors work?

Recall our airplanes chapter; the wings are what create lift, and this is heavily dependent on the wings shape. On quadcopters, each rotor is made up of two or more *blades* attached to a central hub, and these blades are specifically designed to push air *opposite* the direction the aircraft needs to go.

[PLACEHOLDER: ISOMETRIC VIEW OF BLADE WITH LABELING]

Each blade has a *root*, which connects to the hub, and a *tip* at its outer end. Along its length, the blade also has a *twist*, where the blade's angle gradually changes from root to tip, similar to the shape of a screw thread.

[PLACEHOLDER: SPLIT FIGURE OF BLADE TWIST COMPARED TO SCREW MECHANISM]

The direction the air is pushed depends on which way the quadcopter goes; reversing the rotation of a rotor reverses the direction of the air pushed, and thus the direction of the quadcopter.

Rotors are typically described by three numbers: their *diameter*, their *pitch*, and their *number of blades*. Typically, you will see something like *10x5 inch*, which refers to the diameter and pitch, and the visual will tell you the number of blades.

The diameter measures the tip to tip distance that a propeller can reach. If there is only 1 blade, this measurement is twice the hub-to-tip distance. Larger diameter rotors move a bigger column of air, but at slower speeds. Thrust scales roughly equal to the square of velocity, while power scales to the cube of velocity. Bigger props are meaningfully more efficient. This is the same reason helicopters use one large slow rotor while quadcopters use several small fast ones.

Two is the typical amount of blades per propeller, and ensure symmetry along the Y-Axis, as well as mass-production capability. Adding blades increases total thrust at a given diameter and RPM, but each additional blade partly flies through the turbulent wake left by the blade ahead of it (this is called *aerodynamic interference*), so you get diminishing thrust returns per blade.

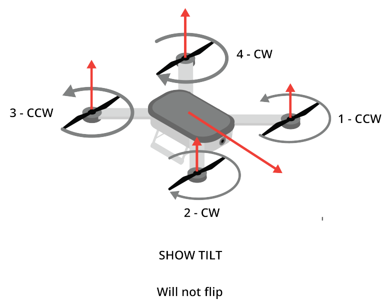
The pitch is the distance the propeller would move after one 360° rotation. Similar to how a screw moves when twisted, this measurement predicts how the propeller would move in vertical space. Lower pitch means less torque needed and higher maximum RPM, but less thrust and vertical movement is produced per rotor. Higher pitch generates more thrust and RPM, but draws more torque from the motor.

5.4 Movement, Spin Direction, and Torque

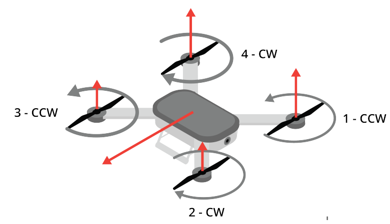
Generally, two rotors spin clockwise (CW) and two spin counter-clockwise (CCW). Rotors positioned diagonally from each other spin in the same direction. Rotors right next to each other spin in opposite directions. If all four rotors spun in the same direction, the body of the drone would violently spin in the opposite direction. By having two CW and two CCW motors, the forces cancel each other out. This balance allows the drone to hover completely still and turn smoothly post-VTOL. There is a guide in your digital resources that goes into blade selection based on your use case, very interesting!

In theory, quadcopters that are symmetrical such as drones could be flown in any direction (this is called *headless flight*). This is easier for beginners as the direction away from the operator is forward, and the direction toward the operator is backwards, regardless of rotation.

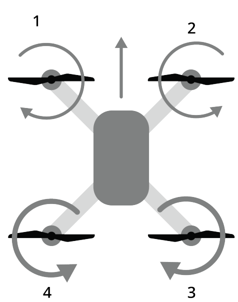
Often a front is decided so that the fuselage body can we can determine roll, pitch, and yaw; this front is called the head. That way, its rotation and flight is based on its relative position in the world, not the operator's location. In the following discussion on forces present on quadcopters, we will assume that there is no drag and our center of gravity is in the center of the fuselage, just as with all other ideal scenarios.



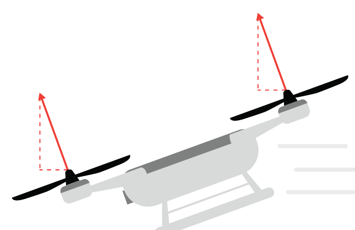
(a) `forwardIso.png`



(b) `lateralIso.png`



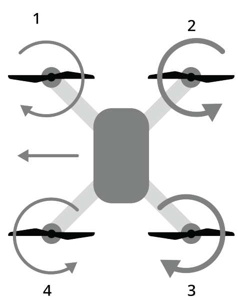
(a) `forwardTop.png`



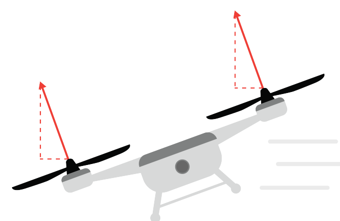
(b) `forwardSide.png`

When flying a quadcopter forward, we apply *pitch* to the quadcopter (figure 5.3a). When doing this, we generate a greater amount of torque from the back two rotors (see 5.4a). This causes the copter to tilt forward with the back rotors higher than the front two rotors (see 5.4b), and direct its motion in the direction of the head. Backwards motion is very similar; the *front* rotors relative to the head spin at a greater rate, and as a result, tilt higher than the back motors, causing backwards motion.

As with airplanes, left and right movement, or *lateral movement*, is controlled by *roll*. Figure 5.3b shows a quadcopter rolling to the left. To produce this roll, the rotors to the *right* of the head spin faster than the rotors to the left, generating extra lift on the right side of the fuselage. This lifts the right side and tips the left side down, tilting the quadcopter's thrust to the left and carrying it in that direction (see 5.5b). Rightward movement works the same way in reverse: the rotors to the *left* spin faster, lifting the left side and rolling the quadcopter to the right.

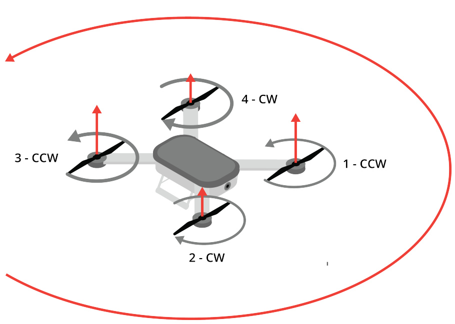


(a) `lateralTop.png`

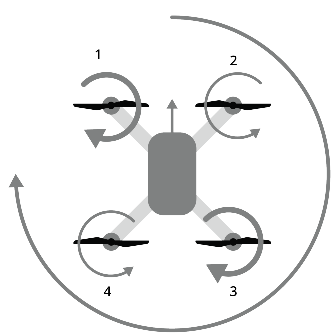


(b) `lateralSide.png`

Rotation of the ahead about the center of gravity is called *yaw*. Instead of two adjacent rotors having increased speeds, we adjust two diagonal rotors to have greater speeds then the other two. Increasing the clockwise rotors will cause a counterclockwise or left rotation. Rightward rotation is similar: increasing the counterclockwise rotors causes a clockwise



(a) `rotationIso.png`



(b) `rotationTop.png`

5.5 Helicopters

Instead of a fixed wing like on an airplane, or four small rotors on a quadcopter, a *helicopter* generates its lift from a single, much larger *main rotor*. The physics is the same as before: the spinning blades push air downward, and Newton's Third Law pushes the aircraft up in response. However, a single big rotor introduces two new problems a quadcopter doesn't have to solve.

First, on a quadcopter, half the rotors spin clockwise and half spin counter-clockwise, so their reaction torques cancel out. A helicopter's main rotor only spins one way, so its reaction torque has nothing to cancel it. Without a fix, the fuselage would spin in the opposite direction of the rotor. Most helicopters solve this with a small *tail rotor* that pushes sideways at a proportional speed to counteract the twist (see Figure 5.7).

Second, a quadcopter changes direction by speeding up or slowing down individual rotors. A helicopter's main rotor spins at a nearly constant speed instead; direction and lift are controlled by changing the *pitch* of the blades as they spin, using controls called *collective* and *cyclic*. We'll cover the tail rotor first, then come back to how collective and cyclic control the main rotor.

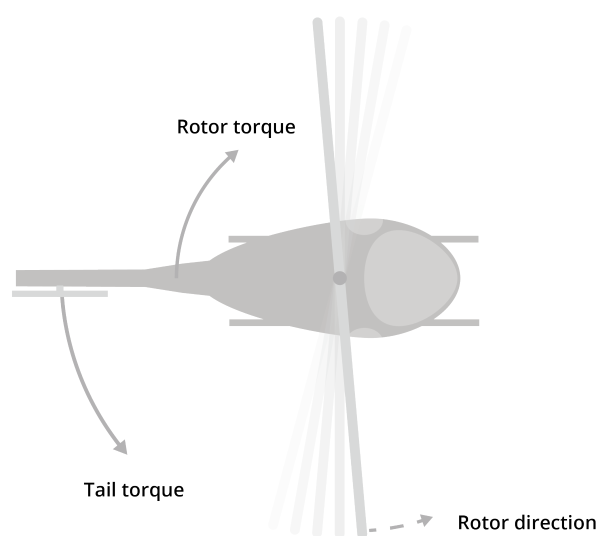
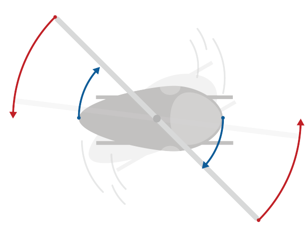


Figure 5.7: A helicopter uses one large rotor that rotates in one direction. A smaller tail rotor counteracts this torque.

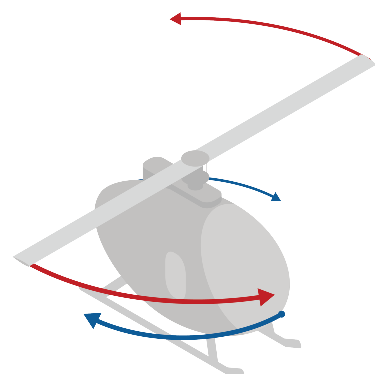
5.5.1 Tail Rotor

On a helicopter, the tail rotor spins vertically, pushing air horizontally to counteract the torque of the main rotor. The pilot controls the tail rotor with pedals, which change the pitch of the tail rotor blades. This changes how much air is pushed sideways, and thus how much torque is counteracted.

Let's imagine a helicopter with a main rotor that spins clockwise and *no tail rotor*. Without a tail rotor, the fuselage would have a violent torque applied to it, and would spin counterclockwise. There is nothing to counteract this torque, so the fuselage would spin faster and faster until the helicopter is destroyed or crashes.



(a) A helicopter without a tail rotor experiences a net torque that spins the fuselage in the opposite direction of the main rotor.



(b) The fuselage continues to rotate faster and faster unless a counter-torque is provided.

Figure 5.8: A helicopter becomes unbalanced without a tail rotor.

Because of this, we need a way to balance this torque. We add a rotor in the back of the helicopter that pushes air in the opposite direction the torque from the main rotor (see Figure 5.9).

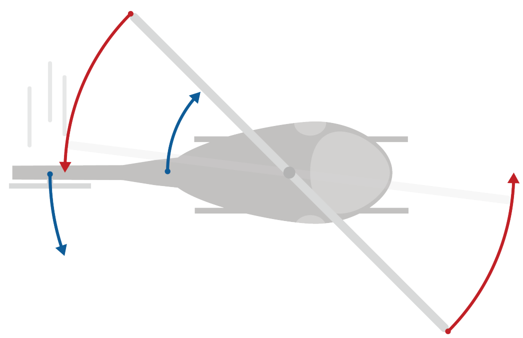


Figure 5.9: A tail rotor provides the counter-torque needed to keep the helicopter balanced and prevent violent movement.

If the pilot wants to yaw left, they increase the pitch of the tail rotor blades, pushing more air to the right and counteracting more of the main rotor's torque. To yaw right, they decrease the pitch of the tail rotor blades, pushing less air to the right and allowing more of the main rotor's torque to spin the fuselage clockwise (see Figure 5.10).

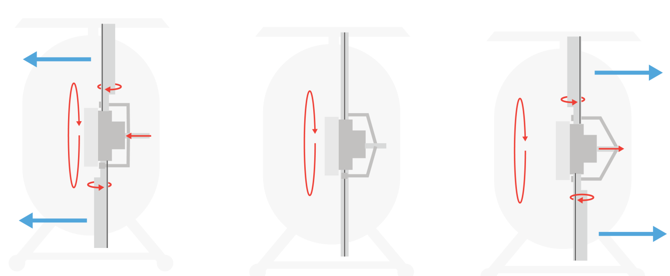


Figure 5.10: The tail rotor changes thrust to control the helicopter's yaw.

5.5.2 Main Rotor Movement

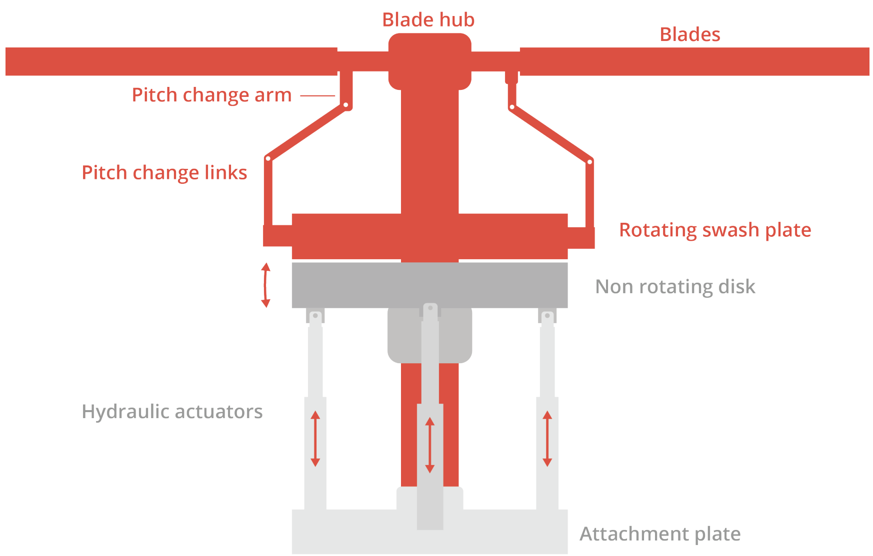


Figure 5.11: `mechanicalControl.png`

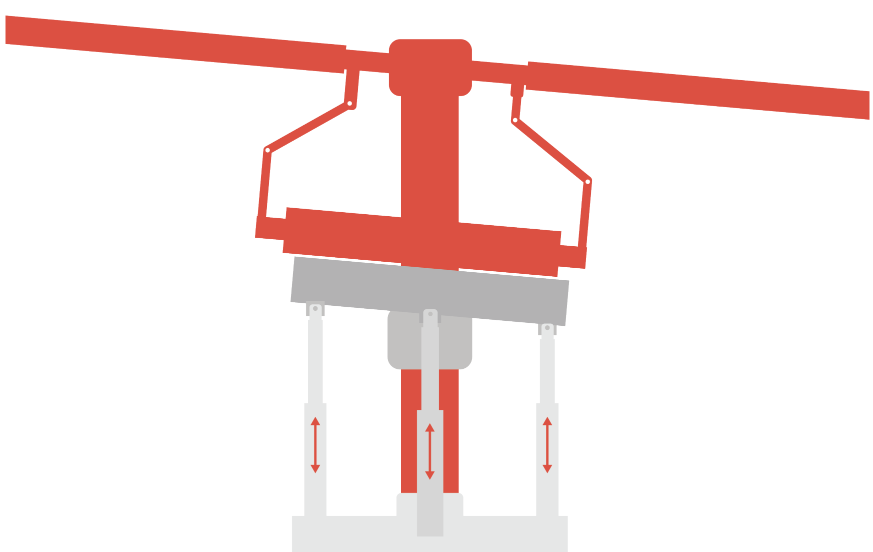


Figure 5.12: `mechanicalControlMovement.png`

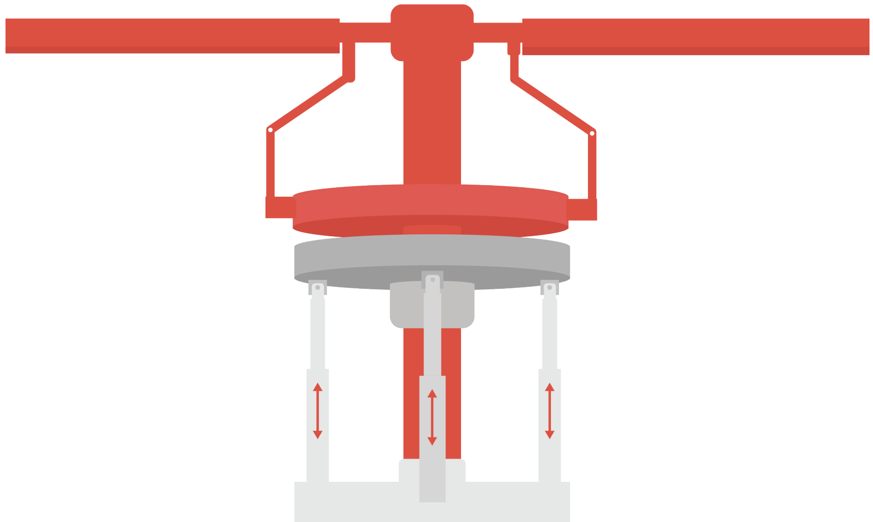
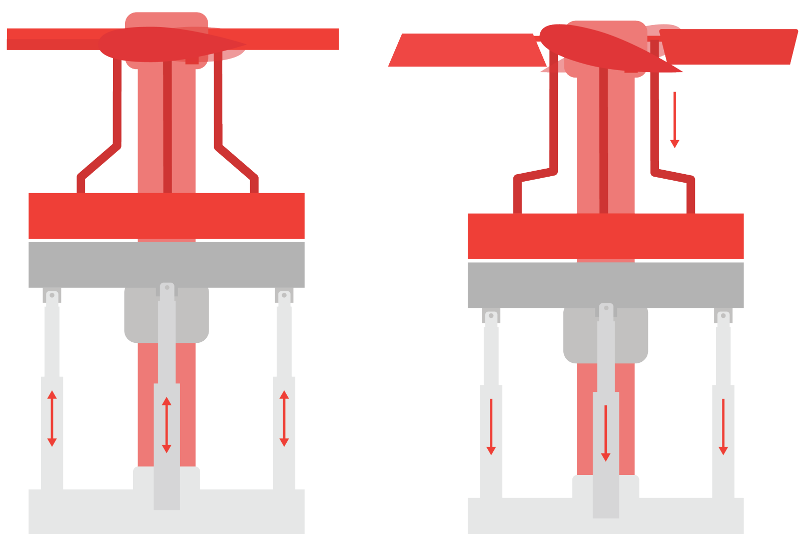


Figure 5.13: `mechanicalControlMovement2.png`



Collective

Figure 5.14: `collective.png`

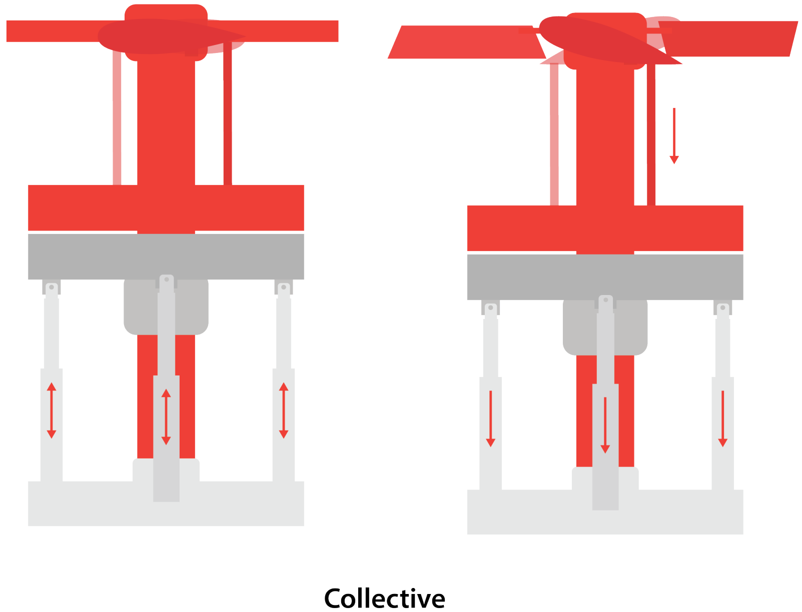


Figure 5.15: `collective4Version.png`

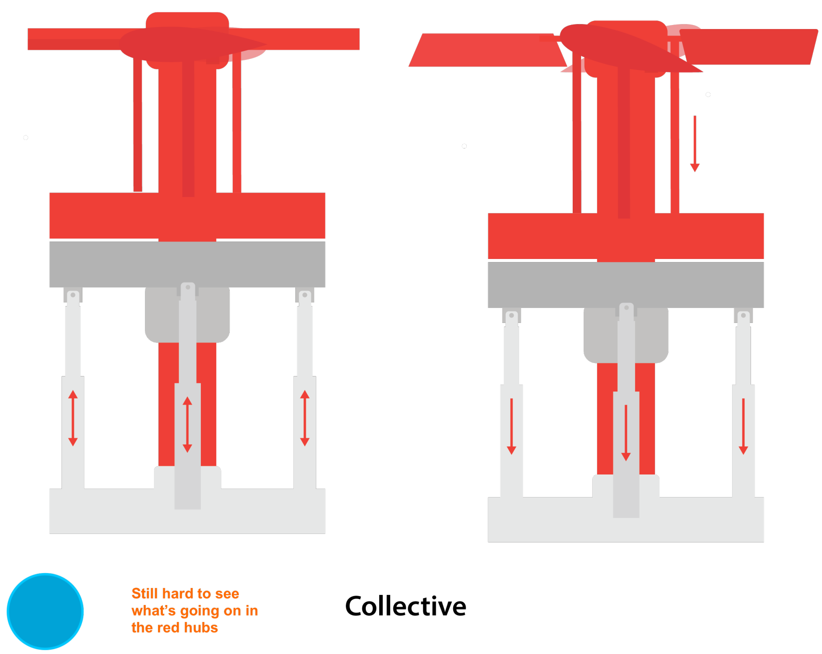


Figure 5.16: `collectiveQuad.png`

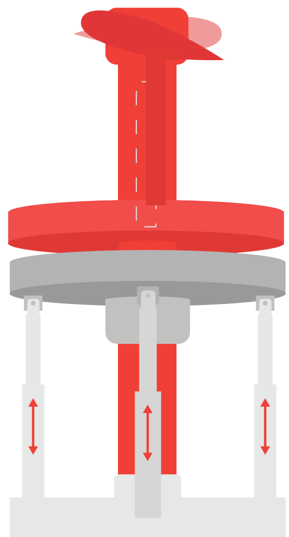


Figure 5.17: `cyclic1.png`

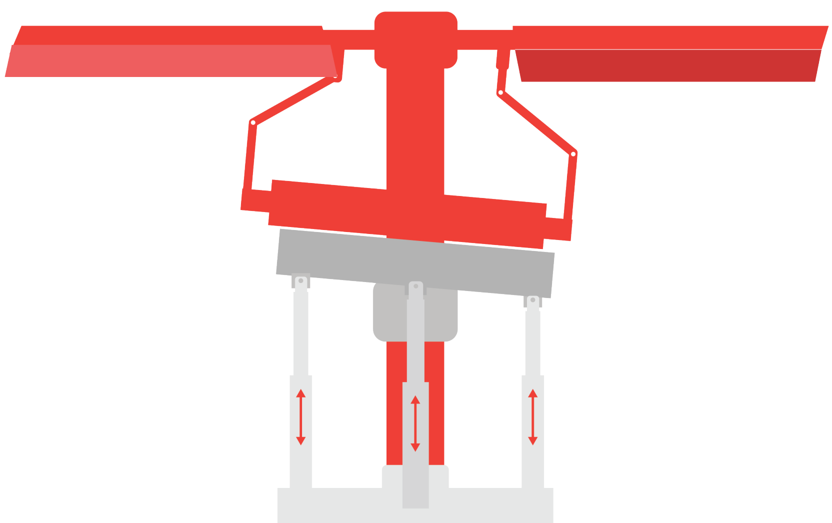


Figure 5.18: `cyclic2.png`

5.5.3 Isometric Copter Views

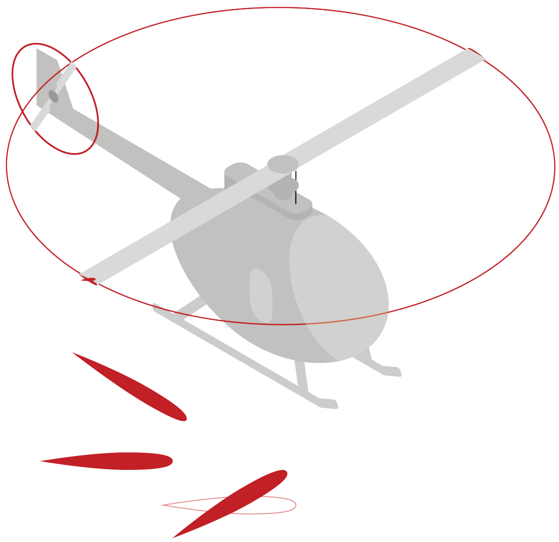


Figure 5.19: `iso-36.png`

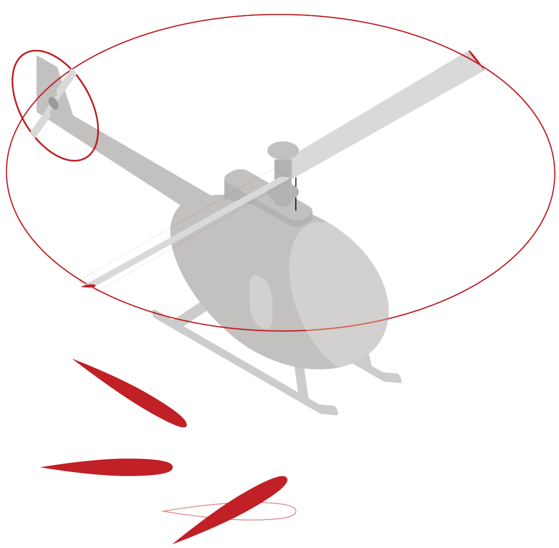


Figure 5.20: `iso-37.png`

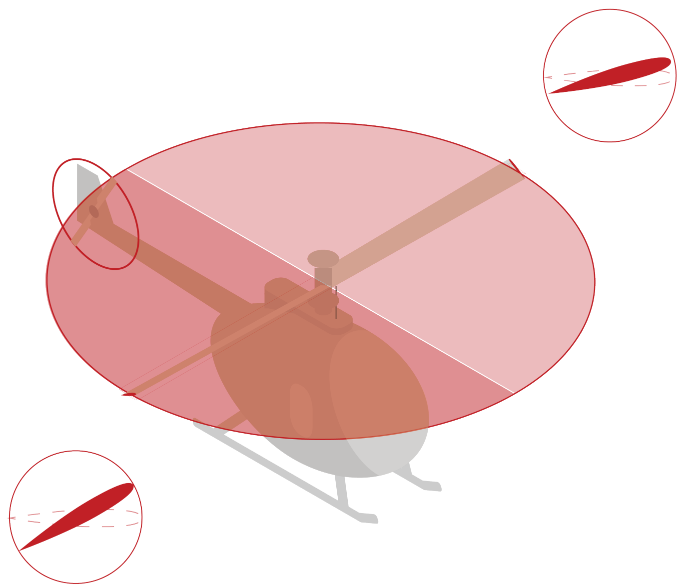


Figure 5.21: iso-38.png

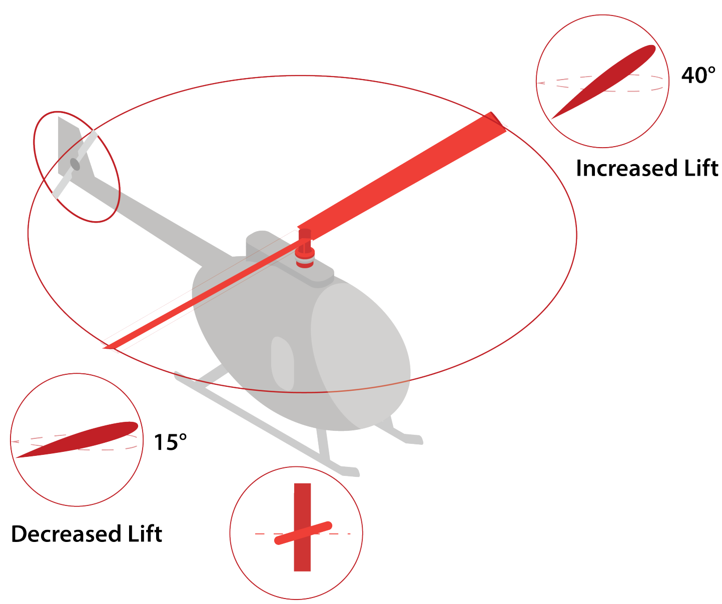


Figure 5.22: iso-48.png

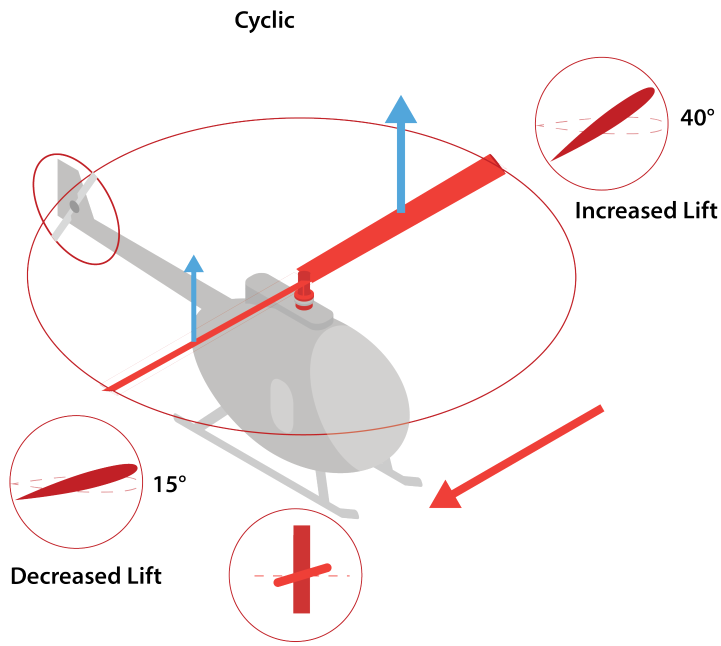


Figure 5.23: `iso-48 2.png`

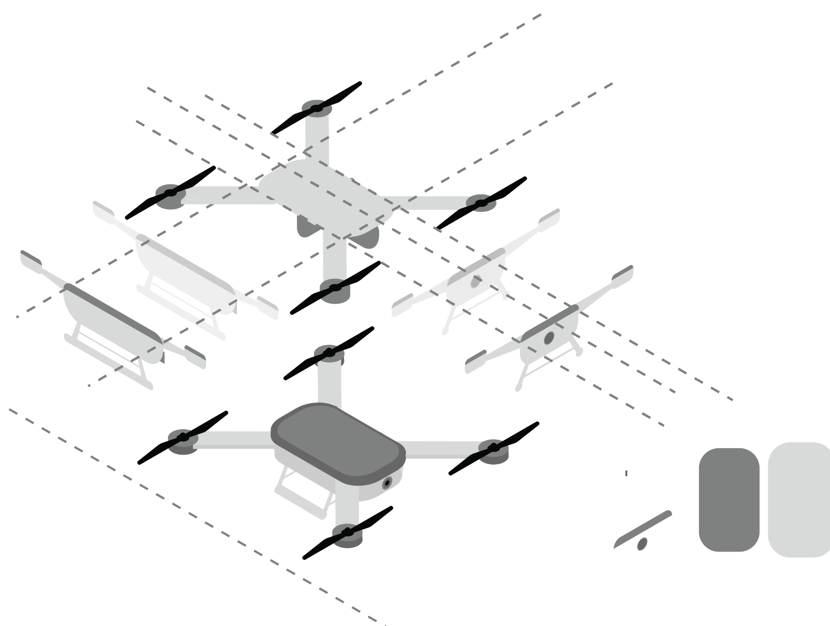


Figure 5.24: `iso.png`

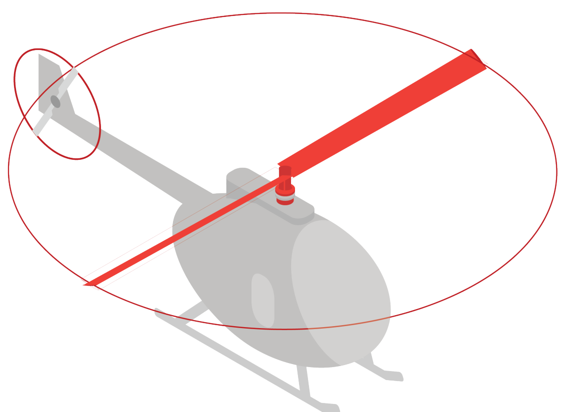


Figure 5.25: `iso46.png`

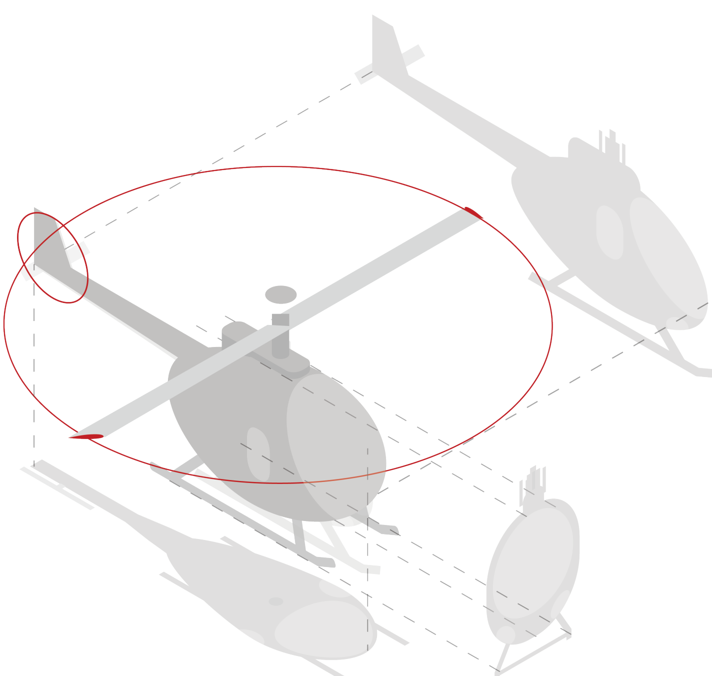


Figure 5.26: `isoprocess.png`

5.5.4 Copter Movement

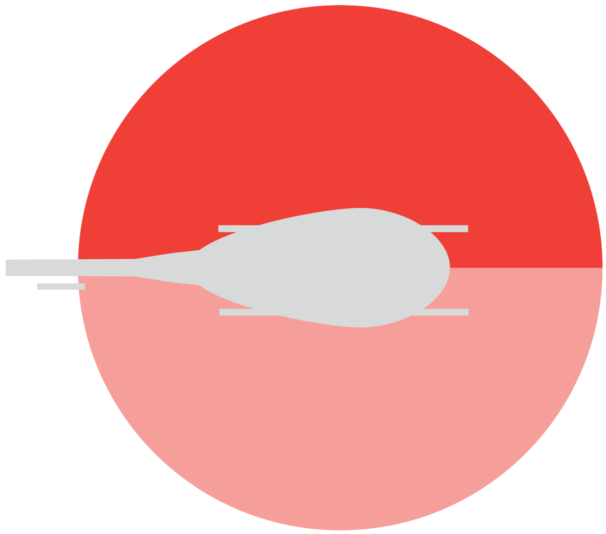


Figure 5.27: `helicopterAbove1.png`

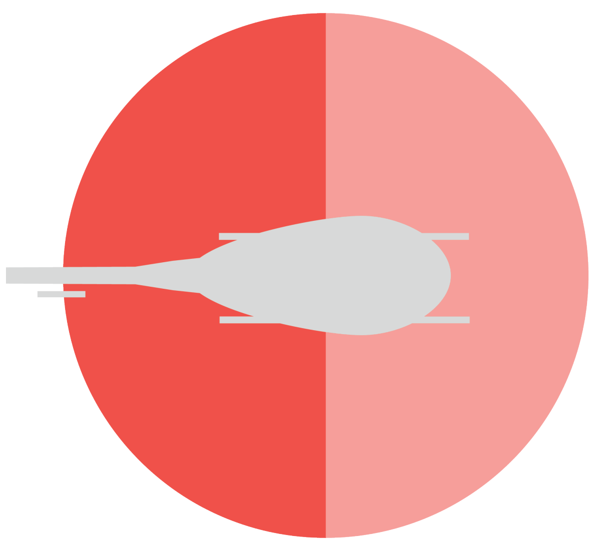


Figure 5.28: `helicopterAbove2.png`

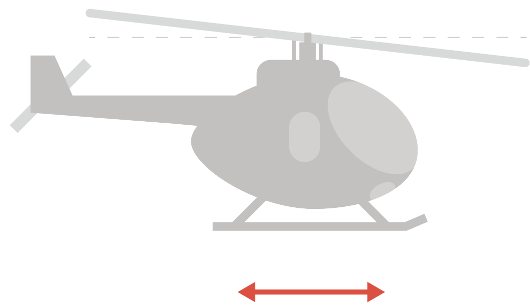


Figure 5.29: `helicopterMovement-10.png`

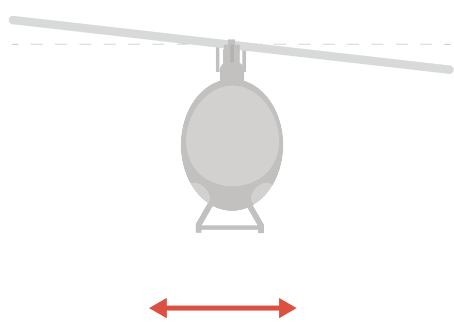


Figure 5.30: `helicopterMovement-11.png`

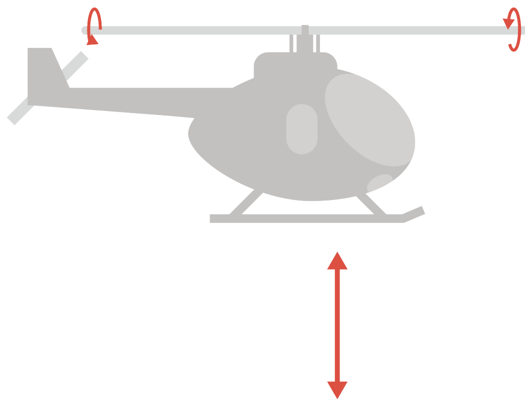


Figure 5.31: `helicopterMovement-12.png`

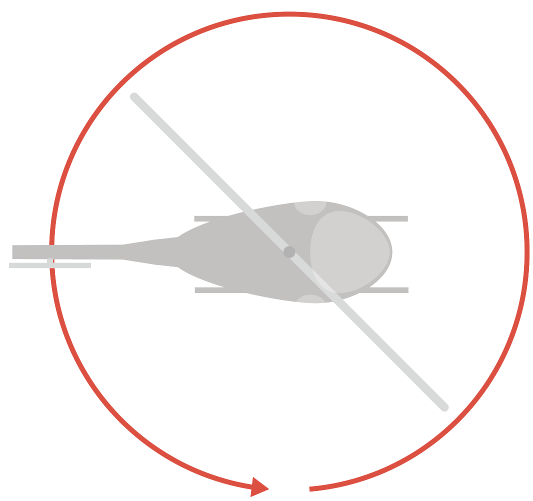


Figure 5.32: `helicopterMovement-13.png`

Appendix A

Answers to Exercises

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